

Substrate or Architecture? A Factorial Evaluation Methodology for Attributing Reliability in Grounded Enterprise Agents

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Abstract. Grounded enterprise agents depend on two things at once: the retrieval design that selects evidence, and the quality of the records that evidence is drawn from. Published evaluations vary the first and hold the second fixed, so a reported reliability figure cannot be attributed to either, although the two causes call for different remedies. In this work we propose an evaluation methodology that separates them. It requires frozen, audited states of one record set, produced under a validation-first discipline in which no transformation is applied before it has been scored on held-out truth, and it scores agent answers against the audited records rather than a hand-written key. We apply the methodology to a hybrid architecture that fuses sparse and dense retrieval and expands the result through a lightweight property graph, using confidential records from a UK manufacturer operating under harmonised European safety standards. Our results demonstrate that graph grounding improves every measured dimension in every data state, most decisively whether a correct answer cites the record it describes. We then show that remediation leaves that success rate largely unchanged while widening the answerable question set by more than a quarter. These findings suggest that retrieval design and record quality contribute to agent reliability independently.

Keywords: hybrid retrieval-augmented generation, knowledge graph, data quality, factorial evaluation, open-weight LLM, traceability, governed AI.

1 Introduction

This paper contributes a way to tell two things apart: how much of a grounded agent's reliability comes from the quality of the records it answers from, and how much comes from the retrieval design that selects them.

Grounded agents are now a common enterprise pattern: a retrieval layer selects evidence from the organisation's own records and a language model is constrained to answer from it. Evaluations of such systems almost always vary the retrieval design over a fixed corpus. That is a reasonable experimental convention, but it makes a reported reliability figure impossible to attribute, because a system can look unreliable for two unrelated reasons. The retrieval may be failing to surface evidence that exists, or the evidence may not be in the records. Those two diagnoses call for entirely different remedies, and an organisation deciding where to spend cannot tell them apart from published results.

Separating them requires something the retrieval literature rarely has: several versions of the same corpus at known, audited levels of quality, so that data quality becomes an experimental variable rather than a fixed background. This paper contributes such an evaluation methodology and applies it. Section 2 covers the retrieval methods and the data-quality discipline the work depends on. Section 3 gives the factorial evaluation methodology and the architecture in generic terms, justified by requirements that hold in any regulated setting. Section 4 instantiates both on an industrial corpus and sets out the evaluation. Sections 5 to 7 report, discuss and bound the findings.

Three research questions organise the study.

RQ1. Holding the records fixed, does graph-grounded hybrid retrieval improve answer accuracy, citation validity and correct abstention over a conventional dense baseline, and is any improvement consistent across data states?

RQ2. Holding the architecture fixed, how does audited remediation of the records change the number of questions that can be answered and the proportion answered correctly, and is that effect larger or smaller than the architecture effect?

RQ3. Does a three-billion-parameter self-hosted model correctly abstain on at least 80% of questions the records cannot support, and does its behaviour hold across all data states without falling back to a non-generative mode?

2 Background and Related Work

2.1 Sparse retrieval and BM25

Sparse lexical retrieval ranks documents by weighted term overlap with the query, and BM25 is the standard formulation [1]. It sums over query terms an inverse document frequency weight multiplied by a saturating term-frequency component, controlled by two parameters: k_1 , which limits how much repeated occurrences of a term can contribute, and b , which normalises for document length against the collection average. The behavioural consequence is that BM25 rewards rare, exact tokens and resists both keyword stuffing and length bias. Records that carry safety and compliance information are full of exactly such tokens, because a standard is referred to by a fixed designation such as EN 388, which is a rare literal string that appears nowhere else. That is why BM25 is a strong baseline in this domain rather than a historical one. Its limitation is equally structural: it matches surface form, so a query worded differently from the record returns nothing, and it has no representation of two designations denoting the same standard.

2.2 Dense retrieval and rank fusion

Dense retrieval embeds queries and passages in a shared space and ranks by similarity, recovering the paraphrase that BM25 misses [2], at the cost of interpretability and with documented sensitivity to noise and distractor passages [3]. Since the two families fail on different inputs, combination is well motivated, and reciprocal rank fusion is the usual combiner because it works on ranks and so requires no calibration between incomparable score scales [4]. Retrieval-augmented generation surveys treat hybrid retrieval as an established strategy [5, 6], while deployment studies find mis-ranked and missing context to be the dominant failure mode in practice [7].

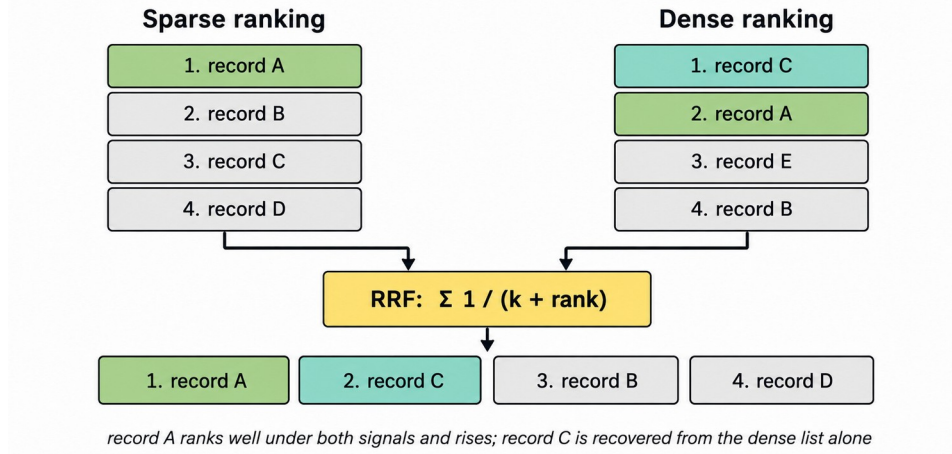


Fig. 1. Reciprocal rank fusion. Two ranked lists produced by incomparable scoring schemes are merged by summing the reciprocal of each rank, so a record ranked well by either signal is promoted and no score calibration is required.

2.3 Knowledge graphs, agents and governance

Labelled property graphs express typed entities and their relations and support traversals that chunk retrieval cannot [8]. Grounding generation on them improves factual accuracy in critical settings [9], and recent graph-augmented methods build entity graphs to answer questions whose evidence is distributed across documents [10, 11], with construction cost, entity resolution and schema maintenance as the recognised barriers [12]. Agent patterns for tool use and stepwise reasoning are established [13, 14, 15], but unverified autonomy is unreliable [16] and models produce fluent unsupported claims [17, 18]. Reference-free evaluation makes groundedness measurable [19],

explainability work treats inspectability as a system property [20], and governance frameworks expect controls to be built into the artefact [21, 22, 23, 24, 25, 26].

2.4 Data quality as an experimental variable

Data quality is conventionally framed through accuracy, completeness, consistency and timeliness [27, 28], with well-documented organisational cost [29] and a persistent gap between how producers and consumers define fitness for use [30]. Cleaning is a workflow rather than a single transformation [31, 32, 33], and entity resolution is central wherever separate systems describe the same physical object [34]. The data-centric AI literature argues that deployed model performance is bounded by data work that is systematically undervalued [35, 36], and production practice extends this to continuous validation of arriving data [37, 38] with per-change documentation [39].

A complementary strand reframes the unit of accountability. Dehghani's data mesh argues that data should be owned and served as a product, with a named owner, a published contract and quality guarantees that consumers can rely on, rather than treated as exhaust from the systems that generate it [40]. That framing is directly relevant here in two ways. It supplies the language for what the audited states are: three versions of a record set published with explicit provenance and measured quality, which is what makes them usable as an experimental variable rather than merely a cleaned table. It also names the failure this paper measures downstream, since an agent consuming records that carry no quality guarantee inherits that uncertainty and passes it to its users as unattributable unreliability.

That literature stops at the cleaned table. It asserts that downstream systems depend on data quality but does not measure the dependency, because doing so requires holding a downstream system fixed while varying the data in a controlled way. The methodology in Section 3 supplies exactly that, which is the connection this paper makes.

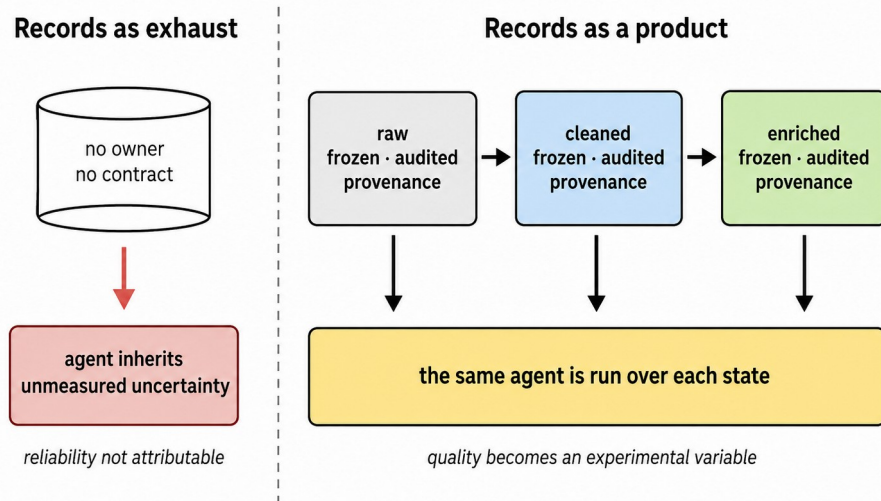


Fig. 2. Records treated as exhaust against records treated as a product. Published, audited states carry provenance and measured quality, which is what allows the same agent to be run over each of them and reliability to be attributed.

2.5 Factorial and decomposition studies of retrieval

A recent line of work asks which component of a retrieval pipeline actually produced a reported gain, rather than reporting the gain alone. Zerhoudi et al. isolate metadata, structure and retrieval strategy across six benchmarks and four models, and show that varying all three at once, which is the usual practice, conceals the fact that most enrichment reduces accuracy [41]. Mishra et al. run a three-by-three matrix of chunking strategies against embedding techniques on enterprise documents and attribute the contribution of each by ablation [42]. Chong reports factorial evidence that document-side, query-side and answer-side interventions are complementary rather than substitutable [43]. Closest to the design used here, Volpini et al. cross three document representations with two retrieval modes, standard retrieval and an agentic mode capable of multi-hop link traversal, over four web domains [44].

Every one of these varies how information is represented, chunked or queried while holding the underlying record set fixed. The factor manipulated in this paper is a different one. The records themselves change, at audited levels of quality, while representation and agent are held constant. Volpini et al. come nearest, since their representations differ

in how much entity structure they expose, but a representation is a rendering of the same underlying facts, whereas a remediated record set contains facts the raw one did not. That difference is what makes the attribution possible, because it separates an architecture able to exploit structure from a substrate that has more to exploit.

2.6 Motivating deployment and gap

The requirements were sharpened by a live deployment with a UK manufacturer and distributor operating under harmonised European safety standards, whose product, specification, supplier and compliance knowledge is spread across an enterprise resource planning system and separate document stores. The records are commercially confidential and cannot be sent to an external model provider, there is no in-house data-science platform, and the costly queries are those that cross entities. Those conditions are not sector-specific. Factorial evaluation of retrieval pipelines is therefore established, and we make no claim to introduce it. What remains insufficiently isolated is the reliability contribution of record quality relative to retrieval architecture, on one confidential enterprise corpus and question set, scored against the audited records rather than against a hand-written key. That is the gap addressed here.

3 Evaluation Methodology and Architecture

3.1 The factorial evaluation methodology

The methodology has three generic requirements.

First, **frozen audited states**. The same record set must exist at two or more quality levels, each immutable and each accompanied by a per-field record of the audited value and its provenance. States must be produced under a validation-first rule: no transformation is applied until it has been scored on held-out truth, and transformations that fail are recorded rather than retried silently.

Audited states, rather than merely different ones, are what make the factorial work, for two reasons. A factorial requires that the levels of a factor be known, and an unaudited cleaning pass gives a corpus that is different by an unmeasured amount, so a difference in agent behaviour cannot be sized against it. More importantly, the audit is what supplies the answer key. Because each state records the defensible value for every field, and marks the fields where none exists, the same instrument can be regenerated per state and applied unchanged. Without that, the key would have to be written by hand once and reused, which would silently penalise a state for holding values the key does not know about. Auditing therefore does double duty: it defines the factor and it measures the outcome.

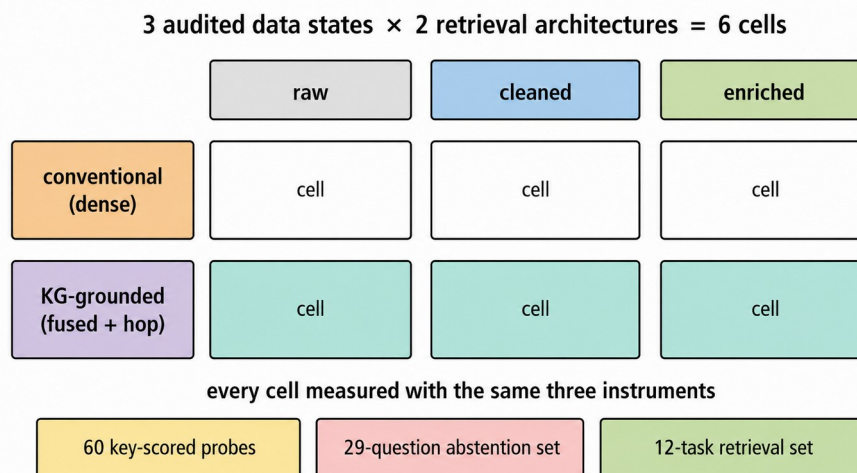


Fig. 3. The factorial design. Three frozen, audited states of one record set are crossed with two retrieval architectures, and every resulting cell is measured with the same three instruments, so an effect can be attributed to the records or to the retrieval design.

Second, **an architecture axis** that can be switched without otherwise changing the system, so that any difference is attributable to retrieval design and not to prompt, model or corpus.

Third, **scoring against the audited records**. Probes are generated from the audited key rather than hand-written, which removes the question of whether the answer key reflects the data, and lets the same instrument be applied unchanged to every state. Crucially, the audited key marks fields for which no defensible value exists, so those probes test correct abstention rather than accuracy. This matters because remediation changes how many probes are answerable, so a rate alone will mislead and coverage must be reported beside it.

3.2 Architecture under test

The agent is a staged pipeline, not an autonomous system, built to three requirements that recur in regulated settings. **Data sovereignty** forces self-hostable components and therefore bounds model size to what commodity hardware serves. **Auditability** requires provenance to be carried end to end, so identifiers assigned at ingestion survive into citations and logs. **Reliability** keeps the generative model out of the control path, selecting among validated options rather than determining behaviour.

Records are indexed three ways over one corpus so tiers are comparable: BM25, dense embeddings, and a labelled property graph whose nodes and edges are drawn from entities already present in the source schema. Building from schema-resident entities rather than inducing a graph from text trades expressiveness for tractability, avoiding the entity-resolution and schema-drift costs identified as the main barrier to industrial graph adoption [12]. The two architectures compared are **conventional**, dense retrieval alone, and **graph-grounded**, which fuses sparse and dense rankings by reciprocal rank fusion [4] and then performs one hop of graph expansion from the top fused results to a shared entity and back to sibling records. The hop count is set to one because deeper traversal grows the candidate set faster than it adds relevant evidence, and added distractors carry a measured cost in retrieval-augmented systems [3].

3.3 Model selection and experimental control

The generative component is interchangeable, because retrieval and traversal are model-independent and grounding is enforced by the prompt and the verification stage. We compare instruction-tuned open-weight models from one family at three and seven billion parameters, so the comparison isolates scale rather than training data or tokeniser, both quantised and served locally through the same interface. The smaller model produces shorter completions and holds less assembled evidence in context, trading claim support for latency. Reporting that trade matters because data sovereignty implies commodity hardware, and the practical question is whether the smaller model is adequate.

For the factorial itself the generator is held fixed, because it is a control rather than a factor. The design already spans three data states and two architectures, and a model axis would multiply cells without addressing RQ1 or RQ2. Holding a model fixed is sound only if it is not so weak that it hides the effects under test, so the model was screened against a threshold fixed in advance (Section 4.3) and met it. The choice is also conservative for both contrasts. Every cell receives the same prompt and the same retrieved evidence, so a stronger generator would be expected to lift all cells rather than reorder them, which means effects visible under the smaller model are unlikely to be artefacts of its weakness. Reporting the compute envelope beside the design follows efficiency-aware reporting guidance [45, 46].

3.4 Supervised tools and verification

An eight-stage workflow runs over shared state: plan, retrieve, graph expand, rank, graph facts, analyse, synthesise, verify. Analysis is restricted to a fixed whitelist of parameterised operations with no arbitrary code execution, and graph queries return the traversal path with the answer. This follows least-privilege guidance for the AI lifecycle [22] and the requirement to treat retrieved content as untrusted, since indirect prompt injection is the leading application-layer risk for retrieval systems [47, 23]; constraining the model to select among validated options also reflects evidence that unverified model plans are unreliable [16]. The limitation is that a whitelist cannot serve a question nobody anticipated, so coverage is bounded by the operations implemented and extending it needs a developer. In a regulated setting a predictable refusal is preferable to an unpredictable action, which is why we accept that ceiling.

4 Implementation and Evaluation

4.1 Producing the audited states

The methodology was instantiated on the product master of Beeswift Limited, a UK manufacturer and distributor of personal protective equipment and workwear, comprising 71,304 records with roughly 100 fields each, spanning two national entities of 50,546 and 20,758 records. Profiling first falsified the organisation's own quality metric: an attribute completeness of about 27% proved to be the web publication rate, since seven fields are populated for exactly the 19,251 published records and for none of the 52,053 unpublished ones. Those fields are captured in the publishing system and never written back, so no diligence at record creation could populate them. A metric can be numerically correct and semantically wrong, which is why states had to be audited rather than asserted.

Remediation separates proposal from application. Seven generation routes emit candidate values with method, confidence and evidence into ledgers: code decomposition with learned colour tokens, base-item inheritance, cross-entity twin comparison, family-sibling recovery, rule-based text extraction, nearest-neighbour imputation under an agreement threshold, and placeholder detection for sentinel values. A master ledger merges 432,692 unique proposals, resolving collisions by trust rank so that relationship-derived evidence outranks deterministic rules, which outrank text extraction. A rule-bound applier then walks the ledger under per-cell invariants: never overwrite a non-blank value, never write a placeholder or zero as a fix, never modify a safety-certified record automatically, and log every change reversibly.

No route counts until validated on held-out truth, and rejections are reported as results. Four plausible methods failed and shipped nowhere (Table 1), preventing roughly 50,000 erroneous writes. The cross-entity estimate is the instructive case: an initial analysis suggested 39,898 cells could be copied where item codes match, and the figure survived an adversarial recount because the recount verified the arithmetic rather than the method. Hold-out testing reduced it to 1,152 safe cells across five fields, since 28 fields are genuinely entity-specific.

Table 1. Recovery methods rejected by held-out validation.

Method	Initial estimate	Validated result	Disposition
Cross-entity twin copy	39,898 cells	1,152 cells safe	Bulk claim refuted
Description-to-fabric extraction	16,760 recoveries	2.8% agreement	Rejected
Cross-family weight regression	22,078 targets	41.1% within 10%	Rejected
Second-pass colour vocabulary	~7,000 targets	92.5%, below 95% bar	Rejected

Three frozen states result: **raw**; **cleaned**, applying 237,496 changes across 68,900 records with no non-blank value overwritten, verified cell by cell; and **enriched**, adding 89,388 individually flagged and reversible fills, with 3,417 withheld because the target record is safety-certified (Table 2). Each state ships with a per-field key giving the audited value and a provenance flag, including a marker for fields where no defensible value exists.

Table 2. Real completeness, placeholders excluded, across states.

Field	Raw	Cleaned	Enriched
Colour	25.6%	73.7%	88.2%
Size	27.0%	79.9%	82.7%
Brand	30.7%	56.1%	78.7%
FabricComposition	9.8%	11.1%	57.3%
Vendor-code	51.7%	51.7%	72.2%

4.2 Agent instantiation and parameters

From each state a bounded corpus of 8,000 records is ingested, selected as those carrying a technical description, so the three corpora contain the same products at different levels of record quality. Ingestion canonicalises standards designations, collapsing surface variants while keeping part numbers distinct, and resolves supplier and brand

spellings by fuzzy clustering with every merge logged. Each record becomes one retrievable unit of at most 900 characters with 120 characters of overlap and a stable identifier that every citation and audit entry references. Three indexes are rebuilt per state over the identical units: BM25, a 384-dimensional sentence-transformer index in a local Chroma collection, and a NetworkX property graph holding 8,388 nodes and 25,769 edges, with 4,598 records carrying at least one of 106 canonical standards. Fusion uses reciprocal rank fusion with the rank constant at 60 and a single expansion hop with a 0.15 score increment for corroborated candidates, returning six units per query in the factorial runs.

4.3 Runtime environment

All six cells run locally on a single commodity laptop with unified memory and no discrete accelerator, which is the envelope data sovereignty implies. Generation is served by a local Ollama runtime using a quantised three-billion-parameter instruction-tuned model at temperature 0.1 with an 8,192-token context. Rebuilding indexes for one state takes under three minutes; a complete cell, comprising the abstention benchmark and the key-scored probes, averages 60 minutes, and the full six-cell design required approximately seven hours of local compute. Every cell was checked for silent degradation to the non-generative fallback and none occurred, so all reported results reflect genuine generation.

4.4 Evaluation design

The design crosses three states with two architectures, holding the model fixed, giving six cells. Three instruments run in every applicable cell.

The audited states are used in three specific ways, and it is worth being precise about them. They are the corpus: each state is ingested and indexed separately, so the three corpora hold the same 8,000 products at different levels of record quality. They are the answer key: the key-scored probes are regenerated from each state's own audited values, so a state is never marked against another state's truth. They are the definition of what counts as answerable: a field the audit marks as having no defensible value produces a probe that tests refusal, not accuracy, and the count of such probes changes with the state. Everything else, the agent, the prompt, the model, the parameters and the instruments themselves, is held identical across all six cells.

Key-scored probes. Sixty probes drawn from the audited key over records present in the corpus, covering colour, size, brand, fabric composition, weight and supplier, each asking for one field of one record. Against the audited value for that state we score answer accuracy, citation validity, and blank refusal where no defensible value exists. Three terms are used throughout and are not interchangeable. A probe is *answerable* in a given state if the audit finds a defensible value for that field, and *unanswerable* if it does not, in which case the correct behaviour is refusal. *Answer accuracy* is the proportion of answerable probes answered correctly. The count of answerable probes is itself a function of state, so accuracy and coverage move independently, which is central to the result.

Abstention benchmark. Twenty-nine questions from Beeswift's real purchasing enquiries, thirteen of which concern pricing, stock, order quantities and lead times that are not ingested, so the correct response is refusal.

Retrieval and security. A twelve-task gold set measures capped recall, precision and reciprocal rank per tier. Its labels are constructed from the corpus, so they are correct within a state but not comparable across states, and retrieval therefore supports the architecture comparison only. An adversarial test inserts a record carrying an injected instruction.

5 Results

5.1 Architecture effect (RQ1)

Graph grounding wins on every measured dimension in every data state (Fig. 4, Table 3). Averaged across states it raises answer accuracy from 0.337 to 0.431, correct abstention from 0.655 to 0.770, capped recall from 0.606 to 0.856 and reciprocal rank from 0.515 to 0.829.

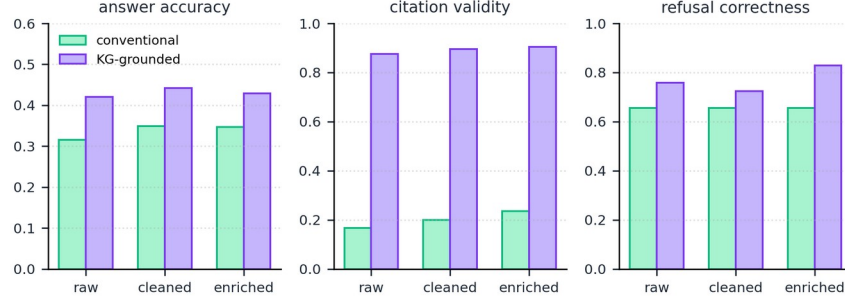


Fig. 4. Architecture effect: conventional dense retrieval against graph-grounded hybrid retrieval, on each data state.

The decisive difference is citation validity, which rises from 0.201 to 0.892. Conventional retrieval often surfaces text supporting a plausible answer without returning the record the question concerns, whereas graph expansion reaches that record and cites it. For an auditable system this matters more than the accuracy difference, because an answer that cannot be tied to a record is not evidence.

Table 3. Combined evaluation matrix. All cells use the three-billion-parameter model; conv = conventional dense, KG = graph-grounded.

State	Arch.	Answer acc.	Citation val.	Abstention	Refused	Recall@10	MRR
raw	conv	0.316	0.167	0.655	10/13	0.658	0.569
raw	KG	0.421	0.875	0.759	11/13	0.883	0.808
cleaned	conv	0.349	0.200	0.655	10/13	0.592	0.517
cleaned	KG	0.442	0.895	0.724	11/13	0.842	0.818
enriched	conv	0.347	0.235	0.655	9/13	0.567	0.458
enriched	KG	0.429	0.905	0.828	12/13	0.842	0.860

5.2 Substrate effect (RQ2)

Read as a rate, answer accuracy is nearly flat across states under graph grounding (0.421, 0.442, 0.429), and reporting only that would be misleading. Remediation does not make the agent better at a fixed set of questions. It enlarges the set that has a defensible answer at all: answerable probes rise from 38 of 60 in the raw state to 43 cleaned and 49 enriched (Fig. 5a).

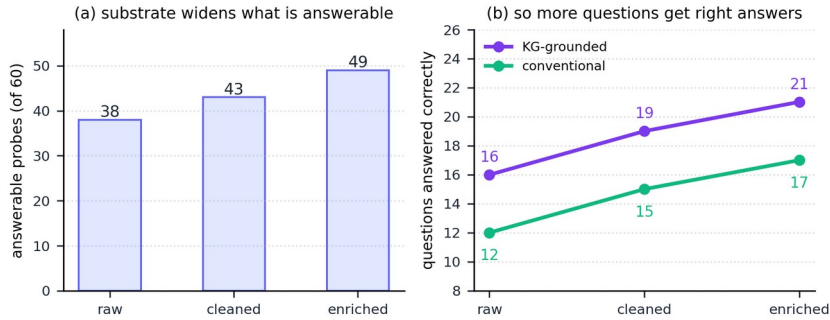


Fig. 5. Substrate effect: (a) remediation widens what is answerable; (b) at a steady success rate that converts into more questions answered correctly.

A steady rate over a growing pool means more correct answers in absolute terms, rising from 16 to 19 to 21 under graph grounding, an increase of 31%, and from 12 to 17 under conventional retrieval, an increase of 42% (Fig. 5b).

Two further measures improve monotonically with the substrate under graph grounding: citation validity from 0.875 to 0.905, and correct abstention from 0.759 to 0.828, reaching 12 of 13 on the enriched state.

Widening the answerable set matters more than it first appears, because a correct refusal is still a question the business did not get answered. An agent that declines because the record is blank has behaved well and been of no use, and the person who asked goes back to reconciling the information by hand, which is the work the agent was deployed to remove. Accuracy rates cannot see this. A system answering a small set of questions perfectly and a system answering a large set equally well score the same, while delivering very different amounts of value. Coverage is therefore not a secondary measure but the one that determines how often the agent is worth consulting at all.

The two effects therefore act on different quantities. Architecture governs the proportion of answerable questions handled correctly and cited properly. Data quality governs the size of the answerable set. Improving retrieval alone raises the hit rate on a small pool; improving data alone widens the pool at an unchanged hit rate. The best cell is enriched records with graph grounding because both quantities move.

5.3 Precision of the estimates

Two properties of the design bound how far these numbers may be pushed, and both are reported rather than left to the reader. Each cell was run once, so no within-cell variance is available, and the abstention denominator is thirteen unanswerable questions, which is small enough that a single changed output moves the proportion by almost eight points.

Table 4. Correct-abstention proportions with 95% Wilson score intervals.

State	Architecture	Correct refusals	Proportion	95% interval
raw	conv	10/13	0.769	[0.497, 0.918]
raw	KG	11/13	0.846	[0.578, 0.957]
cleaned	conv	10/13	0.769	[0.497, 0.918]
cleaned	KG	11/13	0.846	[0.578, 0.957]
enriched	conv	9/13	0.692	[0.424, 0.873]
enriched	KG	12/13	0.923	[0.667, 0.986]

The intervals are wide and they overlap between architectures in every state. The same holds for answer accuracy, where the enriched-state intervals are [0.229, 0.487] for conventional retrieval and [0.300, 0.567] for graph grounding. No single cell in this study distinguishes the two architectures at conventional levels of confidence, and we make no such claim.

What the design does support is a different and weaker form of evidence. Graph grounding is ahead of conventional retrieval on answer accuracy, citation validity, correct abstention, capped recall and reciprocal rank, in all three data states, giving fifteen paired comparisons that agree in direction. Under a null hypothesis of no architectural effect, agreement in that many independent comparisons is unlikely, although the comparisons are not fully independent because they share one corpus and one question set. Consistency of direction across audited states is therefore the claim, and per-cell significance is not.

One quantity in the study is not subject to this caveat. The count of answerable probes rises from 38 to 43 to 49 as the records are remediated, and that is a property of the audited key rather than of a model run: it does not vary with generation, and it is the measurement on which the substrate finding in Section 5.2 rests.

5.4 Model behaviour and security (RQ3)

The threshold set in advance is met, within the precision recorded in Section 5.3. The three-billion-parameter model correctly abstained on 12 of 13 unanswerable questions in the best cell, a proportion of 0.923 with a 95% interval of [0.667, 0.986] against the 80% threshold, and produced no fallbacks to the non-generative mode in any of the six cells, so every result reflects genuine generation. Graph grounding was also the faster architecture in five of six comparisons, because dense-only retrieval carries embedding cost without the sparse index's speed. The injected instruction in the security test was detected in the retrieved text, redacted before prompt assembly and surfaced as a warning, so it never reached the model intact.

6 Discussion

The result reframes a procurement question that is usually posed as a choice. Asking whether to invest in retrieval engineering or in data remediation is a false dichotomy, because the two buy different things. Where the complaint is that answers cannot be verified, that is an architecture problem, and graph grounding moved citation validity from 0.20 to 0.89 here. Where the complaint is that the assistant too often reports that information is unavailable, that is a substrate problem, and remediation moved the answerable share from 63% to 82% of probes. A deployment that fixes only one will improve on one axis and appear stalled on the other.

The design also permits a descriptive reading of the interaction between the two factors, although six cells with one run each cannot test for one. On the two measures that are comparable across states, the architecture advantage narrows as the records improve. The citation-validity gap falls from 0.708 on raw records to 0.670 on enriched, and the answer-accuracy gap from 0.105 to 0.082. In both cases the narrowing comes from the conventional arm rising rather than the graph-grounded arm falling, since graph grounding is already near its ceiling at 0.875 on the raw state. Read carefully, that suggests a small part of what graph grounding buys on poor records is compensation for what those records do not carry, and that remediation recovers a little of it for the weaker architecture. It is not a substitution effect. Conventional citation validity reaches only 0.235 on the best state against 0.905 with graph grounding, so remediation closes a fraction of a gap that remains close to fourfold. Differences of this size also sit within the run-to-run variation a single run cannot bound, so the direction is reported and nothing is inferred from the magnitude.

The methodological point generalises beyond this system. Any reliability figure for a grounded agent is a joint function of design and data, so publishing one without characterising the other leaves it unattributable. The requirements in Section 3.1 are modest, and the instruments are reusable across corpora and architectures.

7 Limitations

The substrate axis is not a pure quality gradient. The three states differ in more than completeness. The enriched state contains values produced by automated methods and flagged as such, so moving from cleaned to enriched changes both how complete the records are and where their contents came from. The comparison is therefore between three defensible operating states of a real catalogue, which is what a deploying organisation actually chooses between, rather than between three points on a single controlled quality scale.

Part of the coverage effect is definitional. Filling fields necessarily increases the number of probes with a defensible answer, so the growth from 38 to 49 answerable probes is partly a consequence of how the audited key is constructed. The result that carries weight is not that coverage grew but that the success rate held as it did, since an agent that answered a widening pool less accurately would have shown a falling rate.

Probe scope. The key-scored probes ask for one field of one record, which is the question type the audited key can adjudicate objectively. That leaves the multi-hop questions, where graph expansion is most valuable, tested only within a single data state by the retrieval instrument, so we cannot report how cross-source recovery behaves as records improve.

Cross-state comparability. Retrieval labels are rebuilt from each state, so retrieval metrics support the architecture comparison but not the substrate comparison, and the apparent fall in dense recall across states is an artefact of changing labels rather than evidence that remediation harms retrieval. The blank-refusal measure rests on a denominator that shrinks from 22 unanswerable probes to 11 as fields are filled, so it is directional only.

Design size and estimate precision. Six cells, one generator and one run per cell support effect directions and sizes rather than significance [46, 48]. Two consequences are quantified in Section 5.3. Without repeated runs no within-cell variance can be estimated, and generation at low temperature is not deterministic, so a single run gives a point estimate whose spread is unknown. And the thirteen unanswerable questions produce confidence intervals roughly thirty points wide, which overlap between architectures in every state. Repeated runs and a larger abstention set are the first two extensions this methodology requires, and neither changes the design: both are parameters of the harness rather than of the method. Two claims are robust to that: the architecture effect on citation validity is large and holds in the same direction in all three states, and the coverage effect is a count that does not vary with generation at all. The smaller differences, particularly abstention correctness across states, are indicative. The generator is held fixed as a control for the reason given in Section 3.3, and the larger model was excluded because it degraded to the non-generative fallback on the hardware available, which is a constraint self-hosting imposes [45].

8 Conclusion

We contributed a factorial evaluation methodology that separates the two dependencies of a grounded enterprise agent, its retrieval design and the quality of its records, and applied it to a traceable hybrid architecture on confidential industrial data. Graph-grounded retrieval beat the conventional dense baseline on every dimension in every data state, most decisively on whether a correct answer cites the record it describes. Audited remediation left the success rate broadly unchanged but widened the answerable set by more than a quarter, converting into 31% more questions answered correctly. Architecture decides how well an agent answers; the records decide how much it can answer. The finding this supports is that reliability is jointly determined by data quality and architecture, so reporting either alone overstates what a deployment will deliver. Both the method and the architecture transfer to any organisation whose records are fragmented, confidential and subject to audit.

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